

MEDIATEK

MT8676 Android General Introduction

Version: 1.0
Release date: 2024-08-12

Use of this document and any information contained therein is subject to the terms and conditions set forth in Exhibit 1. This document is subject to change without notice.

Version History

Version	Date	Author	Description
1.0	2024-08-12	Xinmei Tan	Official release

Table of Contents

Version History 2

Table of Contents 3

List of Figures 3

List of Tables 3

1 MT8676 General Introduction 4

 1.1 MT8676 Overview 4

 1.2 MT8676 Basic Infomation 4

 1.3 MT8676 System Block Diagram 5

 1.4 MT8676 SoC Specification 5

Exhibit 1 Terms and Conditions 7

List of Figures

Figure 1-1. MT8676 system block 5

List of Tables

Table 1-1. MT8676 SoC specification 6

1 MT8676 General Introduction

1.1 MT8676 Overview

The MT8676 is a high-end automotive-grade 5G System-On-Chip (SoC) launched by MediaTek. It is designed for integrated cockpit and parking systems, offering exceptional cost-performance.

The MT8676 device is a highly integrated and scalable application automotive processor with rich multimedia features and AI capabilities. The chip integrates 4 Arm® Makalu and 4 Arm® Klein, along with a powerful multi-standard video codec. Furthermore, an extensive set of interfaces is integrated for camera modules, external audio components, UFS/SD cards, and external modules.

The ARM® Makalu application processor offers substantial computing power to support the latest open-source operating systems, as well as In-Vehicle-Infotainment (IVI) and Cockpit Domain Controller (CDC) applications. The ARM® Klein provide adequate computing power for dedicated tasks, such as software-based graphic rendering and real-time operating system in critical execution environments.

In addition, the SMMU with unique performance enhancements for hard real-time masters, integrated in MT8676, provide a straightforward hardware mechanism that minimizes the virtualization overhead of hypervisors. This is particularly beneficial when multiple operating systems coexist for various application domains, thus optimizing overall system performance.

The new Gen 7 APU in MT8676 is capable to adapt to the latest AI trends in achieving maximizing effective performance in AI-multimedia, AI-camera, and AI-voice experiences. The APU Gen 7 also aims to ensure that AI-enhanced technologies perform sustainably across various conditions for extended durations.

The multi-standard video accelerator and an advanced audio subsystem are also integrated to provide advanced multimedia applications and services such as streaming audio and video, a multitude of decoders and encoders.

The high-performance CPU, DSP, and hardware coprocessors are combined to provide a powerful modem subsystem capable of supporting NR Sub6, LTE Cat 18, Category 24 HSDPA downlink and Category 7 HSUPA uplink data rates, as well as Class 12 GPRS, EDGE.

MT8676 also embodies wireless communication devices, including WLAN, Bluetooth and GPS. With four advanced radio technologies by Combo chipsets, MT8676 provides the best and most convenient connectivity solution in the industry.

The chip supports rich automotive camera features, such as around view monitoring, back view monitoring, automotive driving recording, and driver monitoring. These features can be employed via the maximum 16x video stream inputs with MIPI-CS12, or Ethernet AVB inputs. The rich display interface (MIPI-DSI and DP) allows for support of up to 6 display panels, further enhancing its usability in various automotive applications.

1.2 MT8676 Basic Information

- **System:** Yocto 5.0

- **Kernel:** kernel-6.1
- **Chipset:** MT8676 + MT6197 (RF) + MT6637 (CON) + MT6363 & MT6373 & MT63190 & MT6319U (PMIC)

1.3 MT8676 System Block Diagram

MT8676 system block diagram is shown in Figure 1-1:

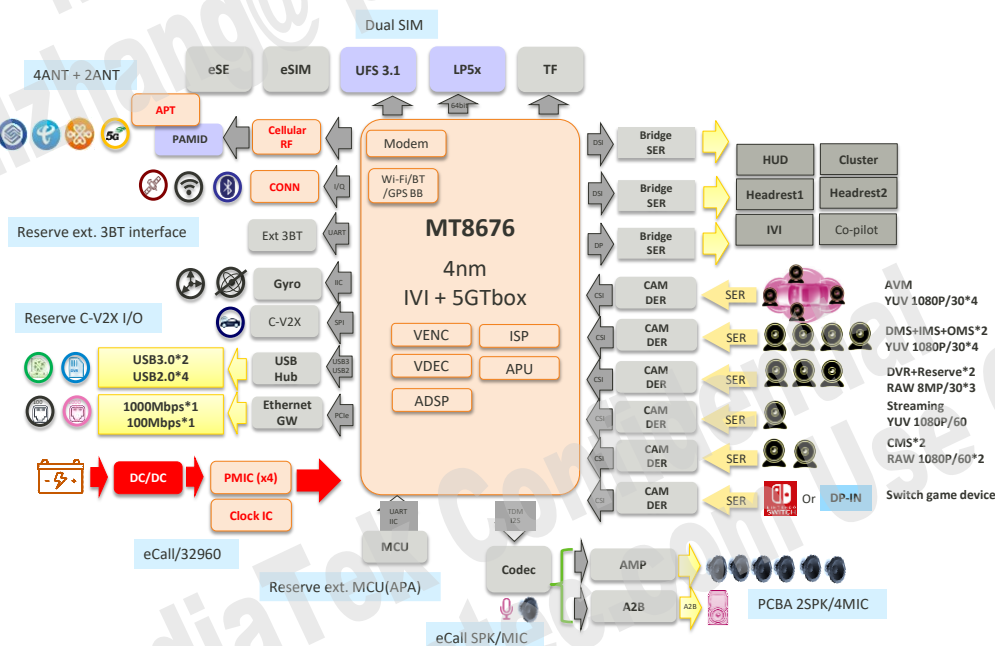


Figure 1-1. MT8676 system block

1.4 MT8676 SoC Specification

MT8676 SoC specification is shown in Table 1-1:

Table 1-1. MT8676 SoC specification

Category	5G SoC MT8676
Process	4nm
CPU	170K DMIPS
GPU	Mali-G615 MC6 1.8T FLOPS
APU	MVPUx1 + MDLAX2, 21.5TOPS
DRAM	LP5x 64bit/7500Mbps, 24GB
Flash	UFS4.0 2lane x1
Camera	19CAM, CSI x6
Display	6 Display DSI 8lane/2.5Gbps + DP1.4 4lane Up to 8K1K
Video Decoder	4K/60, H.264/H.265
Video Encoder	4K/60, H.264/H.265
Audio	ADSP HiFi3*2/800MHz TDM: 8OUT&8IN, I2S: 4OUT&5IN
Modem	5G 3CC R16 & LTE Cat-19
Connectivity	Wi-Fi 6E, BT5.3, GNSS 4M2B/10Hz
Ethernet	Ext PCIe MAC/PHY
PCIe	PCIe3.0 1lane 8Gbps x1
USB	USB3.2 Gen 1 x 1
OS	Yocto 5.0
AECQ & ASIL	AECQ-104, QM/System cluster ASIL-B

Exhibit 1 Terms and Conditions

Your access to and use of this document and the information contained herein (collectively this “Document”) is subject to your (including the corporation or other legal entity you represent, collectively “You”) acceptance of the terms and conditions set forth below (“T&C”). By using, accessing or downloading this Document, You are accepting the T&C and agree to be bound by the T&C. If You don’t agree to the T&C, You may not use this Document and shall immediately destroy any copy thereof.

This Document contains information that is confidential and proprietary to MediaTek Inc. and/or its affiliates (collectively “MediaTek”) or its licensors and is provided solely for Your internal use with MediaTek’s chipset(s) described in this Document and shall not be used for any other purposes (including but not limited to identifying or providing evidence to support any potential patent infringement claim against MediaTek or any of MediaTek’s suppliers and/or direct or indirect customers). Unauthorized use or disclosure of the information contained herein is prohibited. You agree to indemnify MediaTek for any loss or damages suffered by MediaTek for Your unauthorized use or disclosure of this Document, in whole or in part.

MediaTek and its licensors retain titles and all ownership rights in and to this Document and no license (express or implied, by estoppels or otherwise) to any intellectual propriety rights is granted hereunder. This Document is subject to change without further notification. MediaTek does not assume any responsibility arising out of or in connection with any use of, or reliance on, this Document, and specifically disclaims any and all liability, including, without limitation, consequential or incidental damages.

THIS DOCUMENT AND ANY OTHER MATERIALS OR TECHNICAL SUPPORT PROVIDED BY MEDIATEK IN CONNECTION WITH THIS DOCUMENT, IF ANY, ARE PROVIDED “AS IS” WITHOUT WARRANTY OF ANY KIND, WHETHER EXPRESS, IMPLIED, STATUTORY, OR OTHERWISE. MEDIATEK SPECIFICALLY DISCLAIMS ALL WARRANTIES OF MERCHANTABILITY, NON-INFRINGEMENT, FITNESS FOR A PARTICULAR PURPOSE, COMPLETENESS OR ACCURACY AND ALL WARRANTIES ARISING OUT OF TRADE USAGE OR OUT OF A COURSE OF DEALING OR COURSE OF PERFORMANCE. MEDIATEK SHALL NOT BE RESPONSIBLE FOR ANY MEDIATEK DELIVERABLES MADE TO MEET YOUR SPECIFICATIONS OR TO CONFORM TO A PARTICULAR STANDARD OR OPEN FORUM.

Without limiting the generality of the foregoing, MediaTek makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does MediaTek assume any liability arising out of the application or use of any product, circuit or software. You agree that You are solely responsible for the designing, validating and testing Your product incorporating MediaTek’s product and ensure such product meets applicable standards and any safety, security or other requirements.

The above T&C and all acts in connection with the T&C or this Document shall be governed, construed and interpreted in accordance with the laws of Taiwan, without giving effect to the principles of conflicts of law.